

Linear Algebra I — Week 3

Span, Linear Independence & Setting Up Linear Systems

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Linear combination and span. A linear combination of $v_1, \dots, v_k \in V$ is $a_1v_1 + \dots + a_kv_k$ with $a_i \in \mathbb{F}$. $\text{span}\{v_1, \dots, v_k\}$ is the set of all of them; $\text{span} \emptyset = \{\mathbf{0}\}$. It is a subspace, and the smallest subspace containing $v_1, \dots, v_k$.

Setting up the system. To test $w \in \text{span}\{v_1, \dots, v_k\}$ in $\mathbb{F}^n$, solve $a_1v_1 + \dots + a_kv_k = w$. Coordinatewise this is n equations in k unknowns, with augmented matrix whose columns are $v_1, \dots, v_k \mid w$. Solution exists $\iff w \in \text{span}$.

Linear independence. $v_1, \dots, v_k$ are *independent* if $a_1v_1 + \dots + a_kv_k = \mathbf{0}$ forces $a_1 = \dots = a_k = 0$; otherwise *dependent*. Test: solve the homogeneous system; only $\mathbf{0} \Rightarrow$ independent, a nonzero solution $\Rightarrow$ dependent (and it gives a relation).

Useful facts. A set containing $\mathbf{0}$ is dependent. One vector is independent iff it is nonzero. Two vectors are independent iff neither is a scalar multiple of the other. $v_1, \dots, v_k$ dependent $\iff$ some $v_j \in \text{span}$ of the others. If $w \in \text{span}\{v_1, \dots, v_k\}$, adding w does not enlarge the span. Independence and span depend on the field: $\{(1, i), (i, -1)\}$ is dependent over $\mathbb{C}$ but independent over $\mathbb{R}$.

Counting. More homogeneous unknowns than equations forces a nonzero solution; so more than n vectors in $\mathbb{F}^n$ are dependent.

Worked Examples

Example 1 · Membership in a span

Is $(4, 7) \in \text{span}\{(1, 2), (3, 5)\}$ in $\mathbb{R}^2$?

Solution.

1. We need numbers a, b with $a(1, 2) + b(3, 5) = (4, 7)$.
2. Expand: $a(1, 2) = (a, 2a)$ and $b(3, 5) = (3b, 5b)$. Add: $(a + 3b, 2a + 5b)$.
3. Match coordinates: Eq. 1: $a + 3b = 4$; Eq. 2: $2a + 5b = 7$.
4. From Eq. 1: $a = 4 - 3b$.
5. Substitute into Eq. 2: $2(4 - 3b) + 5b = 7$, i.e. $8 - 6b + 5b = 7$, i.e. $8 - b = 7$, so $b = 1$.
6. Then $a = 4 - 3(1) = 1$.
7. Check: $1(1, 2) + 1(3, 5) = (1, 2) + (3, 5) = (4, 7) \checkmark$.

Answer: Yes: $(4, 7) = 1(1, 2) + 1(3, 5)$.

Example 2 · Membership (positive)

Is $(1, 2, 3) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. We seek a, b with $a(1, 0, 1) + b(0, 1, 1) = (1, 2, 3)$.
2. Expand: $a(1, 0, 1) = (a, 0, a)$ and $b(0, 1, 1) = (0, b, b)$. Add: $(a, b, a + b)$.
3. Match coordinates: $a = 1$, $b = 2$, $a + b = 3$.
4. The first two equations give the values $a = 1$ and $b = 2$ directly.
5. There are three equations for two unknowns, so check the third: $a + b = 1 + 2 = 3$, which equals the required 3 $\checkmark$.
6. All three equations hold, so the vector is in the span.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$.

Example 3 · Membership (negative)

Is $(1, 2, 4) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. As in the previous example, $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$.
2. Matching $(1, 2, 4)$ requires $a = 1$, $b = 2$, $a + b = 4$.
3. The first two equations force $a = 1$ and $b = 2$.
4. Test the third: $a + b = 1 + 2 = 3$, but the equation requires 4. Since $3 \neq 4$, the system is inconsistent: no a, b satisfy all three equations.
5. So $(1, 2, 4)$ is not a linear combination of the two vectors.

Answer: No.

Example 4 · Dependence, finding a relation

Test $\{(1, 2, 1), (2, 1, 0), (1, -1, -1)\}$ for independence.

Solution.

1. Suppose $a(1, 2, 1) + b(2, 1, 0) + c(1, -1, -1) = \mathbf{0}$.
2. Expand coordinatewise: $a(1, 2, 1) = (a, 2a, a)$, $b(2, 1, 0) = (2b, b, 0)$, $c(1, -1, -1) = (c, -c, -c)$.
Sum: $(a + 2b + c, 2a + b - c, a - c)$.
3. Set each coordinate to 0: Eq. 1: $a + 2b + c = 0$; Eq. 2: $2a + b - c = 0$; Eq. 3: $a - c = 0$.
4. Eq. 3 gives $c = a$.
5. Substitute $c = a$ into Eq. 1: $a + 2b + a = 2a + 2b = 0$, so $b = -a$.
6. Substitute $b = -a$ and $c = a$ into Eq. 2: $2a + (-a) - a = 0$, true for every a . So Eq. 2 adds no new restriction and a is a free variable.
7. Choose $a = 1$ (any nonzero value): $(a, b, c) = (1, -1, 1)$, a nontrivial solution.
8. Check: $v_1 - v_2 + v_3 = (1, 2, 1) - (2, 1, 0) + (1, -1, -1) = (1 - 2 + 1, 2 - 1 - 1, 1 - 0 - 1) = (0, 0, 0)$
✓.

Answer: Linearly dependent: $v_1 - v_2 + v_3 = \mathbf{0}$.

Example 5 · Independence in P_2

Show $\{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution.

1. Suppose $a(1 + x) + b(x + x^2) + c(1 + x^2)$ is the zero polynomial.
2. Expand and collect powers of x : $a + ax + bx + bx^2 + c + cx^2 = (a + c) + (a + b)x + (b + c)x^2$.
3. A polynomial is zero only if all coefficients are 0. Constant term: $a + c = 0$. Coefficient of x : $a + b = 0$. Coefficient of x^2 : $b + c = 0$.
4. From $a + c = 0$: $a = -c$. From $a + b = 0$: $b = -a = c$.
5. Substitute into $b + c = 0$: $c + c = 2c = 0$, so $c = 0$.
6. Then $a = -c = 0$ and $b = c = 0$. Only the trivial solution exists, so the set is linearly independent.

Answer: Linearly independent.

Example 6 · The field matters

Is $\{(1, i), (i, -1)\}$ linearly independent in $\mathbb{C}^2$ over $\mathbb{C}$? Over $\mathbb{R}$?

Solution.

1. Write $v_1 = (1, i)$ and $v_2 = (i, -1)$. First look for a complex multiple: $i \cdot v_1 = (i \cdot 1, i \cdot i) = (i, i^2) = (i, -1) = v_2$ (using $i^2 = -1$).
2. Over $\mathbb{C}$: $i v_1 - 1 \cdot v_2 = \mathbf{0}$ is a relation with coefficients i and -1 , not both zero. So the set is dependent over $\mathbb{C}$.
3. Over $\mathbb{R}$: suppose $a(1, i) + b(i, -1) = (0, 0)$ with real a, b . Expand: $(a + bi, ai - b) = (0, 0)$.
4. First coordinate $a + bi = 0$: a complex number is 0 only when both real and imaginary parts are 0, so $a = 0$ and $b = 0$.
5. Only the trivial solution exists over $\mathbb{R}$, so the set is independent over $\mathbb{R}$.

Answer: Dependent over $\mathbb{C}$; independent over $\mathbb{R}$.

Example 7 · Proof: dependence means one vector is redundant

Prove: if $v_1, \dots, v_k$ are linearly dependent, then some v_j is a linear combination of the others.

Solution.

1. By the definition of dependence there are scalars $a_1, \dots, a_k$, *not all zero*, with $a_1 v_1 + \dots + a_k v_k = \mathbf{0}$.
2. Since not all the a_i are zero, pick an index j with $a_j \neq 0$.
3. Isolate the j -th term: move all other terms to the right side, $a_j v_j = -\sum_{i \neq j} a_i v_i$.
4. Divide both sides by a_j . This is legal because $a_j \neq 0$ and every nonzero element of a field has an inverse: $v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i$.
5. The right side is a linear combination of the vectors v_i with $i \neq j$, so v_j lies in their span.

Answer: $v_j \in \text{span}\{v_i : i \neq j\}$.

Example 8 · Describing a span by an equation

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

Solution.

1. We need numbers x, y with $x(1, 1, 0) + y(0, 1, 1) = (a, b, c)$.
2. Expand: $x(1, 1, 0) = (x, x, 0)$ and $y(0, 1, 1) = (0, y, y)$. Add: $(x, x + y, y)$.
3. Match coordinates: $x = a, x + y = b, y = c$.
4. The first and third equations give $x = a$ and $y = c$.
5. Substitute into the middle equation: $a + c = b$, i.e. $a - b + c = 0$.
6. Converse: if $a - b + c = 0$ then $b = a + c$ and $a(1, 1, 0) + c(0, 1, 1) = (a, a + c, c) = (a, b, c)$. So the condition is exactly right.

Answer: $a - b + c = 0$

Practice Problems

Part A — Linear Combinations and Span

1. [Foundational]

Write $(5, -1)$ as a linear combination of $(1, 1)$ and $(1, -1)$ in $\mathbb{R}^2$.

Hint. Write $(5, -1) = a(1, 1) + b(1, -1)$ and compare the two coordinates. You get two equations in a and b ; add them, then subtract them.

2. [Foundational]

Decide whether $(2, 3, 4) \in \text{span}\{(1, 1, 1), (0, 1, 2)\}$ in $\mathbb{R}^3$. If so, give the coefficients.

Hint. Set $(2, 3, 4) = a(1, 1, 1) + b(0, 1, 2)$ and compare coordinates. The first coordinate gives a ; then check that the second and third coordinates agree on one value of b .

3. [Foundational]

Decide whether $(1, 0, 0) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Hint. Set $(1, 0, 0) = a(1, 1, 0) + b(0, 1, 1)$. The first coordinate fixes a and the third fixes b ; then test the second coordinate with those values.

4. [Intermediate]

Find k such that $(1, 2, k) \in \text{span}\{(1, 1, 0), (1, -1, 2)\}$.

Hint. Write $(1, 2, k) = a(1, 1, 0) + b(1, -1, 2)$. The first two coordinates form a system in a, b : solve it, then the third coordinate gives k .

5.

[Intermediate]

In $\mathbb{C}^2$ over $\mathbb{C}$, write $(1 + i, 2)$ as a linear combination of $(1, i)$ and $(1, -i)$.

Hint. Set $(1 + i, 2) = a(1, i) + b(1, -i)$ with $a, b \in \mathbb{C}$. The coordinates give $a + b = 1 + i$ and $i(a - b) = 2$. Divide the second by i , using $\frac{1}{i} = -i$.

6.

[Foundational]

Show that $\text{span}\{(1, 2), (2, 4)\} \subseteq \mathbb{R}^2$ is a line, and give its equation.

Hint. One vector is a multiple of the other, so the span is the set of all multiples of a single nonzero vector. Write a general element $t(1, 2)$ and find the relation between its x and y .

7.

[Intermediate]

Describe $\text{span}\{(1, 0, 2), (0, 1, -1)\} \subseteq \mathbb{R}^3$ by a single linear equation in x, y, z .

Hint. A general element is $a(1, 0, 2) + b(0, 1, -1) = (a, b, 2a - b)$. Read off $x = a$ and $y = b$, then express z in terms of x and y .

8.

[Challenge]

Prove that for any vectors $v_1, \dots, v_k$ in a vector space V , the set $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Hint. Use the subspace test: the set is non-empty (it contains $\mathbf{0}$) and closed under addition and scalar multiplication. Add two general linear combinations and regroup the terms by v_i .

Part B — Linear Independence and Dependence

9.

[Foundational]

Is $\{(1, 2), (3, 6)\}$ linearly independent in $\mathbb{R}^2$?

Hint. Check whether one vector is a scalar multiple of the other, or solve $a(1, 2) + b(3, 6) = \mathbf{0}$ and look for a solution other than $a = b = 0$.

10.

[Foundational]

Is $\{(1, 2), (2, 3)\}$ linearly independent in $\mathbb{R}^2$?

Hint. Solve $a(1, 2) + b(2, 3) = \mathbf{0}$ coordinate by coordinate. Is $a = b = 0$ the only solution?

11.

[Intermediate]

Is $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$ linearly independent in $\mathbb{R}^3$?

Hint. Solve $a(1, 0, 1) + b(1, 1, 0) + c(0, 1, 1) = \mathbf{0}$. Three equations in a, b, c : add and subtract them to isolate each unknown.

12.

[Intermediate]

Is $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ linearly independent in $\mathbb{R}^3$? If not, give a nontrivial relation.

Hint. Compare $v_2 - v_1$ with $v_3 - v_2$. Rearranging that observation gives a relation among v_1, v_2, v_3 .

13.

[Challenge]

Find all real k for which $\{(1, 1, 1), (1, 2, 3), (1, 3, k)\}$ is linearly dependent.

Hint. Solve $a(1, 1, 1) + b(1, 2, 3) + c(1, 3, k) = \mathbf{0}$. Use the first equation to eliminate a , express b in terms of c , and ask when c can be nonzero.

14.

[Foundational]

In $P_2(\mathbb{R})$, is $\{1 + x, 1 - x, 2\}$ linearly independent?

Hint. Compare coefficients: in $a(1 + x) + b(1 - x) + 2c = 0$ the coefficient of 1 and of x must both vanish. How many equations and how many unknowns do you have?

15.

[Intermediate]

In $\mathbb{C}^2$, consider $\{(1, i), (1 + i, i - 1)\}$. (a) Is it linearly independent over $\mathbb{C}$? (b) Is it linearly independent over $\mathbb{R}$?

Hint. (a) Over $\mathbb{C}$ the scalar may be complex: divide the first coordinates, then test the second. (b) Over $\mathbb{R}$, write $a(1, i) + b(1 + i, i - 1) = \mathbf{0}$ with real a, b and separate real and imaginary parts.

16.

[Challenge]

Let u, v, w be linearly independent in a real vector space. Prove that $u + v$, $v + w$, $u + w$ are linearly independent.

Hint. Start from $a(u + v) + b(v + w) + c(u + w) = \mathbf{0}$. Collect the coefficients of u , v and w , then use the independence of u, v, w .

Part C — Setting Up Systems, Spanning Sets and Proofs

17.

[Foundational]

Set up and solve the system that decides whether $(1, 2, 3) \in \text{span}\{(1, 0, 1), (2, 1, 0), (0, 1, 1)\}$.

Hint. Write $(1, 2, 3) = a(1, 0, 1) + b(2, 1, 0) + c(0, 1, 1)$ as three equations: $a + 2b = 1$, $b + c = 2$, $a + c = 3$. Substitute to eliminate one unknown at a time.

18.

[Intermediate]

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 2, 0), (0, 1, 1)\}$.

Hint. Set $(a, b, c) = s(1, 2, 0) + t(0, 1, 1)$. The first coordinate gives s and the third gives t ; substitute both into the second coordinate.

19.

[Intermediate]

Find a finite spanning set for $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

Hint. Solve the condition for one variable, for example $x = -y - z$, and write a general vector with y, z free. Split it into a part that contains y and a part that contains z .

20.

[Intermediate]

(a) Show that $\{(1, 1), (1, 2), (2, 1)\}$ spans $\mathbb{R}^2$. (b) Is it linearly independent?

Hint. (a) For a general (x, y) , solve $a(1, 1) + b(1, 2) = (x, y)$ using only the first two vectors. (b) Compare the number of vectors with the dimension of $\mathbb{R}^2$, or find a relation among the three.

21.

[Intermediate]

Show that $\{1, 1 + x, 1 + x + x^2\}$ spans $P_2(\mathbb{R})$.

Hint. Take an arbitrary $a + bx + cx^2$ and write it as $p \cdot 1 + q(1 + x) + r(1 + x + x^2)$. Match the x^2 coefficient first, then x , then the constant.

22.

[Challenge]

Show that $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

Hint. Write a general matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ as $\alpha A_1 + \beta A_2 + \gamma A_3 + \delta A_4$ and compare entries: four equations in four unknowns. For independence, set the matrix equal to 0.

23.

[Challenge]

Prove: if $w \in \text{span}\{v_1, \dots, v_k\}$ then $\text{span}\{v_1, \dots, v_k, w\} = \text{span}\{v_1, \dots, v_k\}$.

Hint. Prove two inclusions. $\text{span}\{v_1, \dots, v_k\} \subseteq \text{span}\{v_1, \dots, v_k, w\}$ is immediate; for the reverse, write $w = \sum c_i v_i$ and substitute it into a general combination.

24.

[Challenge]

Let $v_1, \dots, v_k$ be linearly independent and $v_{k+1} \notin \text{span}\{v_1, \dots, v_k\}$. Prove that $v_1, \dots, v_{k+1}$ are linearly independent.

Hint. Suppose $a_1 v_1 + \dots + a_{k+1} v_{k+1} = \mathbf{0}$. Ask what happens if $a_{k+1} \neq 0$ (where would v_{k+1} lie?), then use the independence of $v_1, \dots, v_k$.